Circle the correct answer for each question. If you're unsure, you can write numbers 1, 2, 3, etc. next to the choices to get partial credit.

1. Suppose a test of significance is conducted with $H_0 : \pi = 0.50$ and $H_A : \pi \neq 0.50$ with a resulting p-value of 0.07. Also, 99% and 95% confidence intervals are constructed from the data. What will be true about the confidence intervals?

   (a) The 99% interval will not contain 0.50, but the 95% interval will contain 0.50.
   (b) Both the 99% and the 95% intervals will contain 0.50.
   (c) The 99% interval will contain 0.50, but the 95% interval will not contain 0.50.
   (d) Neither the 99% nor the 95% intervals will contain 0.50.

   Correct answers: (d)

2. Let π be some population proportion of interest and suppose a 99% confidence interval for π is calculated to be (0.32, 0.41) and a 95% confidence interval is calculated to be (0.34, 0.39). Also, suppose the hypothesis test is $H_0 : \pi = 0.33$ and $H_A : \pi \neq 0.33$. What can be said about the corresponding p-value for this test?

   (a) The corresponding p-value will be less than 0.01.
   (b) The corresponding p-value will be between 0.05 and 0.10.
   (c) The corresponding p-value will be greater than 0.10.
   (d) The corresponding p-value will be between 0.01 and 0.05.

   Correct answers: (d)

3. Commercial fisherman working certain parts of the Atlantic Ocean sometimes find efforts being hindered by the presence of whales. Ideally, the whales could be scared away without frightening the fish. Experience has shown that whales leave the sight of a fishing boat 40% of the time. One of the strategies being experimented with is to transmit underwater sounds of a killer whale. On the 52 occasions that this technique has been tried, the technique worked 24 times (that is, the whales immediately left the area). A 99% confidence interval for π is $(0.29, 0.65)$. How would you interpret this interval?

   (a) If another sample of 52 occasions is collected, a 99% chance exists that the sample proportion of whales who leave the sight of a fishing boat will be between 0.29 and 0.65.
   (b) When underwater sounds of killer whales are transmitted, fishermen can be 99% confident that whales leave the sight of a fishing boat between 29% and 65% of the time.
   (c) 99% of whales will leave the sight of a fishing boat if the whales hear sounds of a killer whale.
   (d) When underwater sounds of killer whales are transmitted, fishermen can be are 99% confident that whales leave the sight of a fishing boat 46% of the time.

   Correct answers: (b)

4. Suppose a confidence interval is constructed using repeated tests of significance to develop an interval of plausible values. Two-sided tests are used each time with the following null hypotheses and resulting p-values.

Null	p-value	Null	p-value
$\pi = 0.45$	0.009	$\pi = 0.53$	0.787
$\pi = 0.46$	0.032	$\pi = 0.54$	0.572
$\pi = 0.47$	0.062	$\pi = 0.55$	0.373
$\pi = 0.48$	0.126	$\pi = 0.56$	0.142
$\pi = 0.49$	0.371	$\pi = 0.57$	0.077
$\pi = 0.50$	0.598	$\pi = 0.58$	0.042
$\pi = 0.51$	0.733	$\pi = 0.59$	0.021
$\pi = 0.52$	0.986	$\pi = 0.60$	0.003

Using the results, give a 95% confidence interval.

(a) (0.45, 0.60)
(b) (0.50, 0.54)
(c) (0.46, 0.58)
(d) (0.47, 0.57)

Correct answers: (d)